# C31 Organic Synthesis – Knowledge Organiser

*AQA A-level Chemistry • Set 1: Reagents and Conditions 1* **Paper 2**

| # | Question | Answer |
|---|---|---|
| 1 | Alkene → alcohol (industrial) | Steam, $H_3PO_4$, 300 °C, 6 MPa |
| 2 | Alcohol → alkene | Conc $H_2SO_4$ or $H_3PO_4$, heat |
| 3 | Halogenoalkane → alcohol | NaOH(aq), warm |
| 4 | Halogenoalkane → nitrile | KCN in ethanol/water, reflux |
| 5 | Nitrile → amine | $LiAlH_4$ or $H_2$/Ni |
| 6 | Primary alcohol → aldehyde | Acidified $K_2Cr_2O_7$, distil |
| 7 | Primary alcohol → carboxylic acid | Acidified $K_2Cr_2O_7$, reflux |
| 8 | Aldehyde or ketone → alcohol | $NaBH_4$ |
| 9 | Alkene → bromoalkane | HBr |
| 10 | Benzene → nitrobenzene | Conc $HNO_3$ + conc $H_2SO_4$, below 55 °C |

# C31 Organic Synthesis – Knowledge Organiser

*AQA A-level Chemistry • Set 2: Reagents and Conditions 2* **Paper 2**

| # | Question | Answer |
|---|---|---|
| 1 | Nitrobenzene → phenylamine | Sn + conc HCl |
| 2 | Benzene → phenyl ketone | Acyl chloride + $AlCl_3$ |
| 3 | Alkane → chloroalkane | $Cl_2$, UV light |
| 4 | Halogenoalkane → alkene | KOH in ethanol, hot |
| 5 | Halogenoalkane → amine | Excess $NH_3$ in ethanol, heat, sealed tube |
| 6 | Aldehyde or ketone → hydroxynitrile | KCN, then dilute acid |
| 7 | Carboxylic acid + alcohol → ester | Conc $H_2SO_4$ catalyst, warm |
| 8 | Acyl chloride → ester | Alcohol, room temperature |
| 9 | Acyl chloride → amide | $NH_3$ or amine |
| 10 | Alkene → dibromoalkane | $Br_2$ |